

7. IMPLIED VOLATILITY & NONLINEAR EQUATIONS

Black-Scholes call and implied volatility
 $\tau = T - t > 0$. $c = SN(d_1) - Ke^{-r\tau}N(d_2)$;
 $d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau}$, $d_2 = d_1 - \sigma\sqrt{\tau}$; $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\xi^2/2} d\xi$.
Put-call parity: $c + Ke^{-r\tau} = p + S$.
Require $S > 0$, $K > 0$, $\sigma > 0$, $\tau > 0$; model assumptions in Topic 10.
Implied volatility: $f(\sigma) = c(\sigma) - c_{\text{mkt}} = 0$, $f'(\sigma) = SN'(d_1)\sqrt{\tau} > 0$.
Existence and uniqueness when $\max(S - Ke^{-r\tau}, 0) \leq c_{\text{mkt}} \leq S$.
 $f(0^+) = \max(S - Ke^{-r\tau}, 0) - c_{\text{mkt}}$, $f(\infty) = S - c_{\text{mkt}}$.

Taylor and finite differences
If $f \in C^{n+1}[a, b]$, $x, x+h \in [a, b]$: $f(x+h) = \sum_{k=0}^n \frac{h^k}{k!} f^{(k)}(x) + O(h^{n+1})$.
 $f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) = \frac{f(x) - f(x-h)}{h} + O(h)$ **Require $f \in C^2$ and shifted point in $[a, b]$.**
 $f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$ **Require $f \in C^3$ and $x \pm h \in [a, b]$.**
 $f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2)$ **Require $f \in C^4$ and $x \pm h \in [a, b]$.**

Machine-rounding balance:
 $O(h) + O(\epsilon h) : h \asymp e^{1/2}$, $O(h^2) + O(\epsilon h) : h \asymp e^{1/3}$, $O(h^2) + O(\epsilon h^2) : h \asymp e^{1/4}$.
 h must be representable so $x \pm h$ differs from x .
Scalar iterations
 $e^{(k)} = x^{(k)} - x^*$, $\beta = \lim_{k \rightarrow \infty} \frac{e^{(k+1)}}{(e^{(k)})^p}$; for $v=1$ require $|\beta| < 1$.
Newton: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$. Quadratic locally; **require $f(x^*)=0$, f and f' continuous near x^* , $f' \neq 0$ there, initial guess sufficiently close**; asymptotic ratio $\rightarrow |f''(x^*)|/(2|f'(x^*)|)$.
Secant: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{\frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}}$; order $(1+\sqrt{5})/2$ locally.
Require two distinct initial iterates, nonzero divided differences, starts sufficiently close to a simple root.

Systems / Gauss-Newton
 $r: \mathbb{R}^n \rightarrow \mathbb{R}^m$; $J_{ij} = \frac{\partial r_i}{\partial y_j}$, $r(x+d) = r(x) + J(x)d + O(\|d\|^2)$. For $m=n$,
Newton-Raphson: $J(x^{(k)})d^{(k)} = -r(x^{(k)})$, $x^{(k+1)} = x^{(k)} + d^{(k)}$; **J must be nonsingular at each solve; local quadratic convergence under smoothness and a sufficiently close start.**
For $m \geq n$ nonlinear least squares $\min \|r\|_2^2$: $J^T J d = -J^T r$.
Local step is uniquely defined when J has full column rank.
 $J_{ij} \approx \frac{r_i(x+he_j) - r_i(x)}{h}$, $J_{ij} \approx \frac{r_i(x+he_j) - r_i(x-he_j)}{2h}$; use $h \approx \varepsilon^{1/2}$ forward, $h \approx \varepsilon^{1/3}$ central.

8. NUMERICAL INTEGRATION

Probability integrals and quadrature
PDF p : $p \geq 0$, $\int \mathbb{R} p = 1$. $E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x) dx$, $P(x) = \int_{-\infty}^x p(\xi) d\xi$.
If P is differentiable and strictly increasing, $y = P(x)$ gives $E[f(X)] = \int_0^1 f(P^{-1}(y)) dy$.
 $I(f) = \int_a^b f(x) dx$, $Q(f) = \sum_i w_i f(\xi_i)$, $E = I - Q$. Degree of
precision m : exact for all polynomials degree $\leq m$ and not all of degree $m+1$.
Composite rules
Grid $a = x_0 < \dots < x_N = b$, $h_i = x_i - x_{i-1}$. Trapezoidal: $Q_N^{\text{trap}} = \sum_{i=1}^N \frac{h_i}{2} (f_{i-1} + f_i)$. Uniform $h = (b-a)/N$: $h[1/2 f_0 + \sum_{i=1}^{N-1} f_i + 1/2 f_N]$.
Midpoint: $Q_N^{\text{mid}} = \sum_{i=1}^N h_i f(\frac{x_{i-1} + x_i}{2})$.
 $|E_N^{\text{trap}}| \leq \frac{(b-a)Mh^2}{12}$, $|E_N^{\text{mid}}| \leq \frac{(b-a)Mh^2}{24}$. **Require $f \in C^2[a, b]$ for $O(h^2)$.** Both have degree 1.
Composite Simpson, **uniform grid and even N** :
 $Q_N^S = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \dots + 4f_{N-1} + f_N]$. Degree 3; $|E_N^S| \leq \frac{(b-a)\max\{f^{(4)}\}h^4}{2880}$.
Require $f \in C^4[a, b]$ for $O(h^4)$.
 $|E_{2N}| \approx \frac{|E_N|}{2^2}$, $r \approx \log_2 \frac{|E_N|}{|E_{2N}|}$.

Gauss-Legendre
On $[-1, 1]$, N nodes/weights chosen exact through x^{2N-1} ; degree $2N-1$.
 $N=2$: $\xi = \pm \frac{1}{\sqrt{3}}$, $w=1$; $N=3$: $\xi = 0, \pm \sqrt{\frac{3}{5}}$, $w = \frac{8}{9}, \frac{5}{9}, \frac{5}{9}$.
 $\int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{i=1}^N w_i f(\frac{a+b}{2} + \frac{b-a}{2} \xi_i)$.
Non-smooth and multidimensional
Formal orders are not guaranteed when required derivatives are missing/unbounded. Split at nonsmooth points or change variables to regularise before applying a high-order rule.
Tensor product: $\int_{[0,1]^d} f(x, y) dx dy \approx \sum_{i,j} w_i w_j f(\xi_i, \xi_j)$. A d -fold
($N+1$)-point rule uses $(N+1)^d$ points (**curse of dimensionality**).

9. RANDOM NUMBERS & SIMULATION — DISTRIBUTIONS
True random: physical source. Pseudorandom: deterministic, reproducible, statistically random-like. Quasirandom: deterministic, designed for multidimensional uniformity.
 $X \sim U[a, b]$: $p(x) = \frac{1}{b-a}$, $E[X] = \frac{a+b}{2}$, $\text{Var}(X) = \frac{(b-a)^2}{12}$ (**$a < b$**); if $U \sim U[0, 1]$, $Y = a + (b-a)U$.
 $U \sim U[0, 1]$, $Y = F^{-1}(U) \implies F_Y = F$. **Use a suitable generalised inverse if strict invertibility fails.**
 $Y \sim N(\mu, \sigma^2)$: $p(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi\sigma}}$, $Y = \mu + \sigma Z$ (**$\sigma > 0$**). Independent sums: $\Sigma X_i = \Sigma EX_i$; $\text{Var} \Sigma X_i = \Sigma \text{Var} X_i$. CLT: iid sums are approximately normal for large n and finite variance.

9. RANDOM NUMBERS & SIMULATION — MC/QMC

Monte Carlo
 $E[f] = \int_{\mathbb{R}} f(x)p(x) dx$, $x_i \stackrel{\text{iid}}{\sim} p$
 $E_N[f] = \frac{1}{N} \sum_{i=1}^N f(x_i)$
Estimator is unbiased; RMS error is independent of d .
 $e_N[f] = \frac{\sigma(f)}{\sqrt{N}} = O(N^{-1/2})$
 $e_N[f] \leq \text{tol} \implies N \geq \left(\frac{\sigma(f)}{\text{tol}}\right)^2$
Require independent samples and finite $\text{Var}[f]$. Variance reduction
Control variate: known $\mu_g = E[g]$, estimate $E[f-g] + \mu_g$; useful when $\text{Var}(f-g) \ll \text{Var}(f)$.
Antithetic pairs: **N even**; on symmetric $\mathbb{R}^d$ pair x and $-x$; on $[0, 1]^d$ pair x and $1-x$. Odd component integrates exactly.

Importance sampling: $q(x) = g(x)p(x)$, $E[f] \approx \frac{1}{N} \sum_{i=1}^N \frac{f(y_i)}{g(y_i)}$, $y_i \sim q$.
Require $q \geq 0$, $\int q = 1$, and $g(x) \neq 0$ wherever $f(x) \neq 0$; finite variance of f/g under q .
Quasi-Monte Carlo
Deterministic points:
 $Q_N = \frac{1}{N} \sum_{j=1}^N f(x_j)$, $x_j \in [0, 1]^d$

Koksma-Hlawka:
 $|\int_{[0,1]^d} f - Q_N| \leq V(f) D_N^*$
Requires bounded variation $V(f)$. Star discrepancy:
 $D_N^* = \sup_{y \in [0,1]^d} \left| \prod_{i=1}^d y_i - \frac{\#\{x_j \in [0, y]\}}{N} \right|$, $D_N^* \leq C \frac{(\log N)^d}{N}$
Constant C depends on d .

10. STOCHASTIC INTEGRATION & SDEs — FOUNDATIONS

Random walk / Wiener process
 $t_n = n\Delta t$, $\Delta t = T/N$:
 $\Delta X_n = \mu \Delta t + \sigma(\Delta t)^{\gamma} Z_n$, $E[X_N - X_0] = \mu T$, $\text{Var}(X_N - X_0) = \sigma^2 T(\Delta t)^{2\gamma-1}$; iid $Z_n \sim N(0, 1)$. Finite nonzero limit requires $\gamma = \frac{1}{2}$.
 $S_{n+1} = S_n \left(1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n\right)$.
Wiener W: $W(0)=0$; a.s. continuous paths; increments over non-overlapping intervals are independent: for $a < b \leq c < d$, $W(b) - W(a)$ is independent of $W(d) - W(c)$; $W(t) - W(s) \sim N(0, t-s)$, $W_{j+1} = W_j + \sqrt{\Delta t} Z_j$.
Itô integral / lemma
 $\int_0^T f(W(t), t) dW(t) = \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} f(W(t_n), t_n) [W(t_{n+1}) - W(t_n)]$ (**left endpoints; adapted integrand**).
 $\sum (\Delta W_n)^2 \rightarrow T$, $\int_0^t W dW = \frac{1}{2} W(T)^2 - \frac{1}{2} T$.
 $dX = f(X, t) dt + g(X, t) dW$, $X(0) = X_0$. For $Y = H(X, t)$:
 $dY = (H_t + \frac{1}{2} g^2 H_{xx}) dt + H_x dX$. **Require H continuously differentiable in t and twice continuously differentiable in x .**

10. STOCHASTIC INTEGRATION & BLACK-SCHOLES

Geometric Brownian motion
 $dS/S = \mu dt + \sigma dW$, **$S_0 > 0$, $\sigma > 0$** . $S(t) = S_0 \exp\left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W(t)\right]$;
 $\ln\left(\frac{S(t)}{S_0}\right) \sim N\left((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t\right)$; $S(t)$ is lognormal and positive.
Black-Scholes PDE and conditions
 $V_t + rSV_x + \frac{1}{2}\sigma^2 S^2 V_{xx} = rV$, $S > 0$, $0 < t \leq T$. Delta hedge uses $f = V_S$ short shares. **Assumptions: GBM underlying; constant known r, σ ; no dividends, transaction costs, taxes or arbitrage; continuous trading and short selling allowed.**
 $c(S, T) = \max(S - K, 0)$, $c(0, t) = 0$, $c \sim S - Ke^{-r(T-t)}$, $p(0, t) = Ke^{-r(T-t)}$, $p \rightarrow 0$.
Put-call parity: $S + p = c + Ke^{-r(T-t)}$ (**European options, same K, T**).

Log-price/time transformation
 $y = \ln(S/K)$, $\tau = T - t$, $V(S, t) = U(y, \tau)$:
 $-U_\tau + (r - \frac{1}{2}\sigma^2)U_y + \frac{1}{2}\sigma^2 U_{yy} = rU$; $y \in \mathbb{R}$, $\tau \in [0, T]$.
 $U(y, 0) = K \max(e^y - 1, 0)$, $U \rightarrow 0$ ($y \rightarrow -\infty$), $U \sim Ke^y - Ke^{-r\tau}$ ($y \rightarrow \infty$);
 $U(y, 0) = K \max(1 - e^y, 0)$, $U \sim Ke^{-r\tau}$ ($y \rightarrow -\infty$), $U \rightarrow 0$ ($y \rightarrow \infty$).

11. FDMs FOR BLACK-SCHOLES

Grid, notation and differences
Truncate $S \in [0, S_{\max}]$, **$S_{\max} > K$ and sufficiently large**.
 $\Delta S = \frac{S_{\max}}{N_S + 1}$, $S_j = j\Delta S$, $\Delta t = \frac{T}{N_t}$, $t_n = n\Delta t$. $V_j^n \approx V(S_j, t_n)$.
 $V_t(S_j, t_n) + rS_j V_S(S_j, t_n) + \frac{1}{2}\sigma^2 S_j^2 V_{SS}(S_j, t_n) = rV_j^n$.
 $V_t(S_j, t_n) \approx \frac{V_j^n - V_{j-1}^{n-1}}{\Delta t}$ backward, $O(\Delta t)$.
 $V_S(S_j, t_n) \approx \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S}$, $V_{SS}(S_j, t_n) \approx \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2}$; each $O((\Delta S)^2)$.
Boundary/terminal values from Topic 10.

Backward Euler - EXPLICIT scheme
 $\rho = \frac{\Delta t}{2(\Delta S)^2}$, $\alpha_j = (\sigma^2 S_j^2 - rS_j \Delta S)\rho$, $\beta_j = 1 - r\Delta t - 2\sigma^2 S_j^2 \rho$, $\gamma_j = (\sigma^2 S_j^2 + rS_j \Delta S)\rho$.
 $V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n$. Accuracy $O(\Delta t + (\Delta S)^2)$ when stable.
Sufficient stability: $\sigma^2 \geq r$, $\frac{\Delta t}{(\Delta S)^2} \leq \min_{1 \leq j \leq N_S} \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2}$; then
 $\alpha, \beta, \gamma \geq 0$ and $|\alpha| + |\beta| + |\gamma| \leq 1$. Under this mesh relation overall order is $O((\Delta S)^2)$.
Forward Euler - IMPLICIT scheme
 $\alpha_j = \frac{j\Delta t}{2}(r - \sigma^2)$, $\beta_j = 1 + r\Delta t + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2}$, $\gamma_j = -\frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} - \frac{rS_j \Delta t}{2\Delta S}$. Solve
tridiagonal $\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1}$ with boundary terms
in RHS, backwards $n = N_t - 1, \dots, 0$. Accuracy $O(\Delta t + (\Delta S)^2)$; unconditionally stable. **$\sigma^2 \geq r$ is sufficient for strict diagonal dominance/nonsingularity, not necessary.**
Factor once if r, σ time-independent; total $O(N_t N_S)$.